

Contact

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(LinkedIn)

Top Skills

Data Structures

Algorithms

TypeScript

Languages

English (Native or Bilingual)

Arabic (Native or Bilingual)

Spanish (Professional Working)

Certifications

Certificate of Completion | Stanford
Science Small Groups

Certificate Of Completion | Stanford
University's CCOP

Certificate Of Completion | Stanford
University's Code in Place

Google IT Automation with Python
Using Python to Interact with the
Operating System

Honors-Awards

Dean's List

Summa Cum Laude (4.0) 3x
Computer Science, Physics,
Mathematics

Publications

LeRAAT: LLM-Enabled Real-Time
Aviation Advisory Tool

Heba Alazzehe

CS @ UC Berkeley | AI Research @ Stanford Intelligent Systems
Lab & Stanford Mineral-X | Prev @ Uber & Snapchat | ColorStack |
RTC | GWC

Burlingame, California, United States

Summary

Hi, my name is Heba, and I am a Computer Science student at UC Berkeley. I am passionate about building accessible, impactful technology and using software to solve real-world problems. I am driven by a desire to build technology that empowers people and communities, grounded in hands-on technical experience and collaborative problem-solving.

Experience

Stanford Intelligent Systems Laboratory

AI Researcher

June 2024 - Present (1 year 9 months)

Stanford, California, United States

- Developed an LLM-powered framework using a modular RAG pipeline with aircraft data to enhance AI-driven decision-making systems within sponsored research by Airbus. Presented to Airbus, NASA, Stanford Center for AI Safety, Stanford REU Symposium, and Stanford researchers; currently undergoing in-flight testing sponsored by the U.S. Air Force Test Pilot School.

- Collaborated with leading AI and domain experts, including mentorship from Prof. Mykel Kochenderfer and Prof. Jef Caers, as well as Dr. Kiana Jafari, Dr. Duncan Eddy, Dr. Mansur Arief, and Dr. Ed Katz.

- Co-authored a research paper accepted to the European Conference on Artificial Intelligence (ECAI) 2025.

- Publication: arXiv:2503.16477

- The Stanford Intelligent Systems Laboratory (SISL) at Stanford University focuses on developing advanced algorithms and analytical methods for the design of robust decision-making systems. This includes both manned and

autonomous systems such as aircraft, spacecraft, vehicles, and other systems where intelligent behavior is highly valued.

Stanford Mineral-X

AI Researcher

June 2024 - Present (1 year 9 months)

Stanford, California, United States

- Engineered a 3D geological visualization tool using Python and Mayavi, enabling interactive exploration of subsurface structures for sustainable mineral resource research.
- Implemented advanced features including dynamic slicing, camera path rendering, and image resolution enhancement to support clear, user-friendly analysis.
- Collaborated closely with Prof. Jef Caers and the Mineral-X team to align technical tools with real-world geoscientific workflows.
- Presented research at the Stanford REU Symposium and the Stanford Doerr School of Sustainability, showcasing technical innovations and societal impact to faculty, peers, and industry professionals.
- Mineral-X focuses on technological innovations to create a resilient & sustainable critical mineral supply chain. We develop new pathways in the upstream mineral supply chain, from mineral exploration to processing, in collaboration with industry partners. We develop novel data science and integrated decision-making under uncertainty across the critical mineral value chain.

Game Developers Conference

ColorStack x Reboot Scholar | Game Developers Conference (2x)

March 2025 - Present (1 year)

San Francisco, California, United States

- Selected as a scholarship recipient to attend the 2025 & 2026 Game Developers Conference (GDC) in San Francisco.
- Engaging in industry networking, professional development, and technical discussions on game development and emerging technologies.

- Connecting with top gaming companies and professionals for mentorship and career opportunities.

Stanford University School of Engineering

Stanford Engineering Research Introductions Scholar (SERIS)

December 2024 - February 2026 (1 year 3 months)

Stanford, California, United States

- Competitively selected 1 of 23 from a highly qualified pool of undergraduate candidates across the U.S. to participate in Stanford University's prestigious SERIS program.
- Engaged with world-class researchers and thought leaders to explore cutting-edge innovations and advancements in engineering.
- Developed critical professional and research skills, including navigating graduate school, securing research opportunities, and excelling in academia and industry.
- Participated in immersive workshops, presentations, and discussions led by Stanford faculty and graduate students.
- Built a strong network of accomplished peers and mentors within the Stanford engineering community.
- Enhanced understanding of advanced research methodologies and their applications to real-world challenges in engineering.

Stiles Hall

Experience Berkeley Transfer Ambassador Program Intern

August 2025 - December 2025 (5 months)

Berkeley, California, United States

- Provided targeted, written feedback on multiple rounds of Personal Insight Question (PIQ) drafts to strengthen clarity, storytelling, and admissions competitiveness.
- Mentored California community college transfer students from historically underrepresented backgrounds through the UC application and transition to UC Berkeley.

- Managed a student caseload using Google Classroom and Google Sheets, tracking progress, deadlines, and engagement.
- Hosted 1:1 advising sessions and virtual drop-in hours via Zoom and Calendly to support application strategy and campus acclimation.
- Collaborated on a semester-end program growth project, researching scalable solutions for digital community-building, outreach, or centralized resource access.
- Advanced the program's mission to demystify UC Berkeley, build transfer student confidence, and improve retention and long-term success.

ColorStack

ColorStack Stacked Up Summit Scholar

July 2025 - August 2025 (2 months)

San Francisco Bay Area

- Awarded scholarship to attend ColorStack's Stacked Up Summit, a selective conference supporting under-represented students in tech.

Snap Inc.

Snap Lens Academy Scholar

May 2025 - August 2025 (4 months)

Santa Monica, California, United States

- Selected as 1 of 15 for Snap's AR/VR Lens Academy; prototyped spatial experiences on Snap Spectacles (AR glasses).
- Built 30+ interactive AR Lenses using modular scripting (Lens Studio API), VFX, and 3D assets achieving 500K+ views and engaging millions of global users.
- Presented live demos to the Snap CEO, Snap CTO, engineers, and leadership, leveraging agile development to enhance code quality, performance, and UX.

Uber

Software Engineer Fellow

December 2024 - August 2025 (9 months)

- Selected as 1 of 50 from 2000+ applicants globally being within the top 3% of applicants for Uber's international Career Prep Program.

- Engaged in rigorous workshops, mentorship sessions, and mock interviews with Uber software engineers to refine algorithm-style interview skills and professional communication.

- Completed advanced technical coursework covering arrays, strings, hashmaps, linked lists, trees, graphs, heaps/priority queues, backtracking, greedy algorithms, and dynamic programming.

Salesforce

Salesforce Futureforce Tech Accelerator Attendee

July 2025 - July 2025 (1 month)

San Francisco, California, United States

- Selected to attend a competitive 3-day accelerator focused on AI, automation, and human-AI collaboration

- Participated in hands-on workshops, engineering panels, and networking sessions with Salesforce technical leaders

BNY

BNY Sophomore Summit Pre-Internship Program Fellow

April 2025 - June 2025 (3 months)

- Selected for a competitive, six-week virtual program designed to develop industry-relevant skills, insights, and experiences.

- Engaged in interactive sessions with senior leaders and networking with current and former interns.

- Tackled real-world challenges and explored BNY's innovative culture to enhance professional growth in the financial services industry.

Break Through Tech

AI Fellow

March 2025 - June 2025 (4 months)

- Selected as one of 1,000 fellows nationwide for a rigorous AI and machine learning program

- Cornell University partnership: Earned a certification upon successful completion of Machine Learning Foundations

- Completed Machine Learning Foundations coursework and AI Studio projects using real-world datasets
- Collaborated with industry experts and peers to develop and deploy ML models
- Engaged in mentorship and career development to prepare for roles in AI and data science

Liberty Mutual Insurance

Women In Technology Summit Attendee

February 2025 - June 2025 (5 months)

- Attended an exclusive summit focused on advancing women in technology, fostering leadership, and promoting diversity in STEM.
- Engaged with industry experts, thought leaders, and engineers on topics such as emerging technologies, innovation, and career development in tech.
- Participated in networking sessions, keynote presentations, and discussions on navigating challenges and opportunities in the tech industry.
- Gained insights into Liberty Mutual's technological initiatives and commitment to empowering women in technology.

MongoDB

Women in Computer Science Summit Attendee

February 2025 - May 2025 (4 months)

- Chosen to attend a competitive summit designed for women in computer science, focused on technical growth, career development, and networking.
- Engaged with industry professionals and engineers to explore advancements in database technologies and software engineering best practices.
- Participated in technical discussions, mentorship sessions, and hands-on workshops to enhance expertise in database systems and distributed computing.

San Mateo County Community College District

Web Services Intern

October 2024 - May 2025 (8 months)

San Mateo, California, United States

- Maintain and update the district website, ensuring functionality and user accessibility.
- Assist in developing new web pages and online forms, leveraging content management systems for efficient workflows.
- Troubleshoot and resolve web-related issues, supporting district-wide web applications and third-party software integration.
- Web Services is a team that supports hundreds of web applications and websites across the district, including in-house custom developed applications and third-party software like Zoom and Canvas.

College of San Mateo

1 year 5 months

Computer Science Club Vice President

May 2024 - May 2025 (1 year 1 month)

CSM AI Club Media / Marketing Manager

April 2024 - May 2025 (1 year 2 months)

- Passionate about AI and its transformative potential, I lead our digital presence to connect, engage, and inspire our community. Through strategic content curation, creative campaigns, and interactive engagement, I foster meaningful connections among AI enthusiasts, students, and industry professionals.

CSM Computer Science Club Media / Marketing Manager

January 2024 - May 2025 (1 year 5 months)

- As the Social Media Manager for Computer Science Club, I spearheaded dynamic digital strategies to amplify our presence and engagement within the tech community. Leveraging my expertise in social media marketing and passion for technology, I curated compelling content, managed platforms, and fostered meaningful connections with students, faculty, and industry professionals. From showcasing student achievements to promoting campus events and initiatives, I championed our mission to inspire, inform, and empower the next generation of computer scientists.

Girls Who Code

CSM Girls Who Code Club President | Founder

April 2024 - May 2025 (1 year 2 months)

- As the President/Founder of the CSM Girls Who Code Club, I lead and oversee activities promoting coding education and empowering young women in technology. Through strategic leadership and mentorship, I cultivate a supportive, inclusive environment for members to develop coding skills, explore STEM careers, and build a strong community.

- Leadership: Provide vision to guide club initiatives aligned with its mission and goals.

Management: Organize meetings, workshops, and events supporting coding education and community building.

- Mentorship: Mentor members, offering guidance and resources to develop coding skills and pursue STEM interests.

- Collaboration: Partner with local organizations to provide industry engagement and internship opportunities.

- Advocacy: Promote diversity and inclusion in STEM, raising awareness of girls' empowerment in technology.

- Outreach: Use social media and events to attract new members and raise awareness of the club's impact.

- Founded CSM Girls Who Code Club, providing a supportive community for girls in technology.

- Organized coding workshops, guest speaker events, and community outreach, expanding coding education.

- Fostering an inclusive culture, resulting in increased membership and engagement among girls in tech.

- Empowered girls to pursue coding and technology, offering support and mentorship.

Built a community fostering belonging and camaraderie among girls in technology at CSM.

- Expand club reach and impact through increased outreach and partnerships. Advocate for diversity and inclusion, creating equitable opportunities for girls in tech.

LinkedIn

LinkedIn [In]spire Day Attendee (2x)

March 2025 - April 2025 (2 months)

- Chosen by LinkedIn's Women in Tech group to attend a selective, in-person professional development event focused on preparing the next generation of technologists.
- Participated in keynotes, AI-powered resume workshops, mock interview demos, and networking sessions with industry leaders and peers.
- Gained insights on navigating tech internships and leveraging AI tools in the application process.

Columbia University

EngAGE Scholar

March 2025 - March 2025 (1 month)

- Accepted into Engineering Achievers in Graduate Education (engAGE) an exclusive program that brings together the brightest young minds in engineering and applied science.
- Participated in a virtual tour of Columbia University's world-class facilities and historical campus.
- Participated in interactive sessions with distinguished faculty, current graduate students, and admissions professionals.
- Gained valuable insights and strategies for crafting a competitive graduate school application and advancing my engineering career.

NVIDIA

NVIDIA GTC Scholar

March 2025 - March 2025 (1 month)

- Selected for a sponsored opportunity to attend NVIDIA GTC, one of the premier AI and technology conferences.
- Engaged with industry leaders, researchers, and engineers to explore advancements in AI, machine learning, and accelerated computing.
- Attended keynotes, research presentations, and networking events, gaining insights into cutting-edge innovations shaping the future of technology.

- Expanded professional network and deepened knowledge of GPU computing, AI frameworks, and industry applications through interactive sessions and discussions.

Harvard WECODE: Women Engineers Code Conference

Harvard WECODE Tech Fellow

January 2025 - February 2025 (2 months)

Harvard, Massachusetts, United States

- Selected for a competitive fellowship recognizing leadership and contributions in technology.

- Participating in the world's largest student-run Women in Tech conference, fostering innovation, collaboration, and empowerment.

- Engaging with industry leaders and attending sessions on cutting-edge technologies and professional development.

Stanford University

Researcher

June 2024 - February 2025 (9 months)

Stanford, California, United States

Stanford Systems Intelligence Laboratory & Stanford Mineral-X

Kodely

1 year 2 months

Software Engineer Intern

June 2024 - January 2025 (8 months)

- As a Software Engineer Intern at Kodely, I participate in the full cycle of development, from conceptualizing and designing the UX/UI to creating responsive, engaging, and interactive website. My role involves close collaboration with cross-functional teams to ensure that the end product not only meets but exceeds user expectations. I have honed my skills in front-end and back-end development, user experience design, and project management, contributing to projects that drive user engagement and satisfaction.

Operations Coordinator | Lead Instructor

December 2023 - January 2025 (1 year 2 months)

Burlingame, California, United States

- Developed and delivered STEM lessons covering UX/UI design, coding (python beginner to advanced), game design, app design, circuit design, and electricity.
- Proficient in a diverse range of STEM subjects ,demonstrating strong foundational knowledge in computer science.
- Facilitated project based learning experiences to enhance practical skills and application of theoretical concepts.
- Adapted teaching method to diverse learning styles, fostering an inclusive and collaborative learning environment.

Grupology

Tutor

December 2022 - August 2024 (1 year 9 months)

Burlingame, California, United States

- Evaluate students' learning styles and provide appropriate techniques for maximum understanding.
- Specialized as an individual tutor by simplifying math concepts while coaching students to think critically to solve problems.
- Tutor students in Math subjects ranging from foundational topics to algebra.
- Prepared course materials such as homework assignments and handouts.

Tessellations School

Software Engineer Intern

December 2023 - March 2024 (4 months)

Cupertino, California, United States

- Developed a software solution using Python and Raspberry Pi to monitor HTTP connections with a web server and detect network connectivity issues. The system periodically checks the status of the network connection and alerts users if a connection to the web server is lost. It utilizes subprocess and re modules in Python to execute system commands and parse network interface data. GPIO pins on the Raspberry Pi are used to indicate network status through LED indicators.

- Developed an automated attendance system using SikuliX and Python to streamline the after-school attendance process. The system leverages SikuliX's image recognition capabilities to automatically capture and record attendance data. Python scripts were utilized to orchestrate the automation process, including image recognition, data logging, and integration with existing school management systems.

Education

University of California, Berkeley

Bachelor of Arts - BA, Computer Science · (May 2025 - May 2027)

UC Berkeley College of Computing, Data Science, and Society

Bachelor of Arts - BA, Computer Science · (May 2025 - May 2027)

UC Berkeley Electrical Engineering & Computer Sciences (EECS)

Bachelor of Arts - BA, Computer Science · (May 2025 - May 2027)

College of San Mateo

Associate of Science - AS, Computer Science, Physics, and Math · (June 2023 - May 2025)

Mills High School

High School Diploma · (August 2021 - June 2023)